

Advanced Statistical Methods II

Lecture VII: MM Algorithm

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First Semester, AY 2026–27

Outline of Today's Lecture

Motivation

Majorization-minimization

Minorization-maximization

Constructing surrogate functions

EM as an MM algorithm

Worked example I: robust regression

Worked example II: a scalar penalty problem

MM for likelihood and GLM-type problems

Choosing good surrogates

A few exercises

Summary

- ▶ EM was introduced as a method for incomplete or latent-data likelihoods.
- ▶ The E-step computed conditional expectations of missing sufficient statistics.
- ▶ The M-step maximized an expected complete-data log-likelihood.
- ▶ The ascent property made the observed-data likelihood nondecreasing.

This lecture takes a broader view: many useful iterative algorithms work by replacing a hard objective by an easier surrogate objective. This is the main idea behind MM algorithms ([Hunter and Lange, 2004](#); [Lange, 2016](#)).

Why do we need another optimization principle?



Suppose we want to solve

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \ell(\theta) \quad \text{or} \quad \hat{\theta} = \arg \min_{\theta \in \Theta} f(\theta).$$

In many statistical problems:

- ▶ the objective is not quadratic;
- ▶ the likelihood contains sums inside logarithms;
- ▶ the objective may be nondifferentiable;
- ▶ direct numerical optimization may be unstable;
- ▶ a closed-form update may not be available for the original objective.

Can we replace the hard objective by an easier objective at each iteration, while still guaranteeing improvement in the original objective?

The central MM idea



At iteration t , instead of optimizing the original objective directly, construct a surrogate objective that is easier to optimize.

Goal	Surrogate relation	Update
Minimize $f(\theta)$	majorize f by $g(\theta \mid \theta^{(t)})$	minimize g
Maximize $\ell(\theta)$	minorize ℓ by $q(\theta \mid \theta^{(t)})$	maximize q

MM means: do not attack the mountain directly. Build a local, easy-to-climb approximation whose improvement forces improvement in the original objective.

Two names, one principle



The letters “MM” are used in two closely related ways.

- ▶ **Majorization-minimization**: for minimizing an objective f .
- ▶ **Minorization-maximization**: for maximizing an objective ℓ .

Same logic, different direction

- ▶ If loss is bad and small is good, build an upper bound and push the upper bound down.
- ▶ If log-likelihood is good and large is good, build a lower bound and push the lower bound up.

The approach is often called *optimization transfer*: transfer the optimization problem to a simpler surrogate problem ([Lange et al., 2000](#)).

Definition: majorization



Let $f : \Theta \rightarrow \mathbb{R}$ be the objective we want to minimize. A function $g(\theta \mid \theta^{(t)})$ is said to **majorize** f at $\theta^{(t)}$ if

$$g(\theta \mid \theta^{(t)}) \geq f(\theta) \quad \text{for all } \theta \in \Theta$$

and

$$g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)}).$$

The first condition says the surrogate lies above the objective. The second says it touches the objective at the current iterate.

The MM update for minimization



Once a majorizer is available, define

$$\theta^{(t+1)} \in \arg \min_{\theta \in \Theta} g(\theta \mid \theta^{(t)}).$$

This update is useful when g is much easier to minimize than f .

- ▶ f may be nonlinear, nondifferentiable, or awkward.
- ▶ g may be quadratic, separable, or have a closed-form minimizer.
- ▶ The surrogate changes from iteration to iteration.

The method is not “approximate optimization” in a vague sense. The majorization relation gives a formal descent guarantee.

Theorem: Descent Property of the MM Algorithm.

Suppose $g(\theta \mid \theta^{(t)})$ majorizes $f(\theta)$ at $\theta^{(t)}$, and suppose

$$\theta^{(t+1)} \in \arg \min_{\theta \in \Theta} g(\theta \mid \theta^{(t)}).$$

Then

$$f(\theta^{(t+1)}) \leq f(\theta^{(t)}).$$

This is the minimization analogue of the EM ascent property.

Proof of descent property



The proof consists of three steps.

1. By majorization,

$$f(\theta^{(t+1)}) \leq g(\theta^{(t+1)} | \theta^{(t)}).$$

2. Since $\theta^{(t+1)}$ minimizes the surrogate,

$$g(\theta^{(t+1)} | \theta^{(t)}) \leq g(\theta^{(t)} | \theta^{(t)}).$$

3. By the touching condition,

$$g(\theta^{(t)} | \theta^{(t)}) = f(\theta^{(t)}).$$

Combining the three displays gives

$$f(\theta^{(t+1)}) \leq f(\theta^{(t)}).$$

At iteration t :

1. The surrogate surface sits above the true objective.
2. At the current point, the surrogate and objective agree.
3. We move to a point where the surrogate is smaller.
4. Since the true objective lies below the surrogate, the true objective cannot increase.

The surrogate is allowed to be crude away from $\theta^{(t)}$, but it must be tight at $\theta^{(t)}$ and must bound the objective in the correct direction.

Definition: minorization



Let $\ell : \Theta \rightarrow \mathbb{R}$ be the objective we want to maximize. A function $q(\theta \mid \theta^{(t)})$ is said to **minorize** ℓ at $\theta^{(t)}$ if

$$q(\theta \mid \theta^{(t)}) \leq \ell(\theta) \quad \text{for all } \theta \in \Theta$$

and

$$q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).$$

For likelihood problems, we usually maximize log-likelihoods. Therefore, the relevant MM version is often minorization-maximization.

The MM update for maximization



Given a minorizer q , define

$$\theta^{(t+1)} \in \arg \max_{\theta \in \Theta} q(\theta \mid \theta^{(t)}).$$

This is especially useful when q has a simpler form than ℓ :

- ▶ separable terms instead of coupled terms;
- ▶ quadratic lower bounds instead of complicated log-likelihoods;
- ▶ complete-data style likelihoods instead of observed-data likelihoods.

This is exactly the kind of logic we saw in EM, although EM constructs its lower bound in a special probabilistic way.

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Proof of ascent property



The proof mirrors the minimization case.

1. By minorization,

$$\ell(\theta^{(t+1)}) \geq q(\theta^{(t+1)} \mid \theta^{(t)}).$$

2. Since $\theta^{(t+1)}$ maximizes the surrogate,

$$q(\theta^{(t+1)} \mid \theta^{(t)}) \geq q(\theta^{(t)} \mid \theta^{(t)}).$$

3. By the touching condition,

$$q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).$$

Hence

$$\boxed{\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})}.$$

Where do surrogates come from?



MM is useful only if we can construct good surrogates. Common tools include:

- ▶ Jensen's inequality;
- ▶ tangent-line inequalities for convex or concave functions;
- ▶ quadratic upper or lower bounds;
- ▶ Cauchy–Schwarz and related inequalities.

The art of MM is to choose a surrogate that is easy enough to optimize but tight enough to converge reasonably fast.

Surrogates from tangent inequalities



A basic source of MM algorithms is the supporting hyperplane property.

If h is concave, then

$$h(u) \leq h(u_0) + h'(u_0)(u - u_0)$$

for appropriate u, u_0 . The tangent line lies above a concave function.

If h is convex, then

$$h(u) \geq h(u_0) + h'(u_0)(u - u_0).$$

The tangent line lies below a convex function.

Interpretation

These inequalities can turn complicated nonlinear terms into linear or quadratic surrogate terms.

Surrogates from quadratic bounds



Suppose f is differentiable and its gradient is Lipschitz with constant L :

$$\|\nabla f(\theta) - \nabla f(\eta)\| \leq L\|\theta - \eta\|.$$

Then

$$f(\theta) \leq f(\theta^{(t)}) + \nabla f(\theta^{(t)})^\top (\theta - \theta^{(t)}) + \frac{L}{2} \|\theta - \theta^{(t)}\|^2.$$

This is a quadratic majorizer.

Minimizing this quadratic majorizer gives the gradient-descent update with step size $1/L$.

Gradient descent as an MM algorithm



The quadratic majorizer is

$$g(\theta \mid \theta^{(t)}) = f(\theta^{(t)}) + \nabla f(\theta^{(t)})^\top (\theta - \theta^{(t)}) + \frac{L}{2} \|\theta - \theta^{(t)}\|^2.$$

Differentiate with respect to θ :

$$\nabla_\theta g(\theta \mid \theta^{(t)}) = \nabla f(\theta^{(t)}) + L(\theta - \theta^{(t)}).$$

Setting this equal to zero gives

$$\theta^{(t+1)} = \theta^{(t)} - \frac{1}{L} \nabla f(\theta^{(t)}).$$

Thus a familiar algorithm can be understood as MM.

Surrogates from Jensen's inequality



Jensen's inequality often removes a log of a sum:

$$\log \left(\sum_{k=1}^K a_k \right) = \log \left(\sum_{k=1}^K w_k \frac{a_k}{w_k} \right) \geq \sum_{k=1}^K w_k \log \left(\frac{a_k}{w_k} \right),$$

where $w_k > 0$ and $\sum_k w_k = 1$.

This is the key inequality behind the EM lower bound for mixture likelihoods.

Equality occurs when

$$w_k \propto a_k.$$

A small Jensen example: mixture log-likelihood



For a finite mixture,

$$\ell(\theta) = \sum_{i=1}^n \log \left\{ \sum_{k=1}^K \pi_k f_k(y_i; \psi_k) \right\}.$$

Introduce weights w_{ik} , with $w_{ik} \geq 0$ and $\sum_k w_{ik} = 1$. Then

$$\log \left\{ \sum_k \pi_k f_k(y_i; \psi_k) \right\} \geq \sum_k w_{ik} \log \left\{ \frac{\pi_k f_k(y_i; \psi_k)}{w_{ik}} \right\}.$$

At $\theta^{(t)}$, equality is achieved by

$$w_{ik} = \tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k \mid Y_i = y_i).$$

This is why EM can be viewed as a special MM algorithm.

Observed log-likelihood & EM



Recall the EM notation from Lecture IV.

- ▶ Observed data: Y_{obs} .
- ▶ Missing data: Y_{mis} .
- ▶ Complete data: $Y_{\text{com}} = (Y_{\text{obs}}, Y_{\text{mis}})$.

The observed-data likelihood is

$$L(\theta; Y_{\text{obs}}) = \int L_c(\theta; Y_{\text{obs}}, y_{\text{mis}}) dy_{\text{mis}}.$$

The EM algorithm constructs a lower bound for

$$\ell(\theta) = \log L(\theta; Y_{\text{obs}}).$$

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}} [\ell_c(\theta; Y_{\text{com}}) \mid Y_{\text{obs}}].$$

One can write

$$\ell(\theta) = Q(\theta \mid \theta^{(t)}) + [\text{a positive quantity not depending on } \theta].$$

EM increases Q , and the decomposition ensures that this produces ascent in the observed-data log-likelihood.

EM versus general MM



Feature	EM	General MM
Surrogate source	conditional expectation	any valid inequality
Typical objective	likelihood	likelihood or loss
Missing data needed?	yes, conceptually	no
Guarantee	likelihood ascent	ascent/descent
Update style	E-step/M-step	construct/optimize surrogate

EM is therefore a particularly important member of the larger MM family, but MM can be used even when there is no natural missing-data interpretation.

Why use MM for robust regression?



Consider LAD regression:

$$\hat{\beta}_{\text{LAD}} = \arg \min_{\beta \in \mathbb{R}^p} \sum_{i=1}^n |y_i - x_i^\top \beta|.$$

This is statistically useful because absolute loss is less sensitive to large residuals than squared loss.

But the objective is nondifferentiable when

$$y_i - x_i^\top \beta = 0.$$

MM can convert the absolute-value objective into a sequence of weighted least-squares problems.

A useful inequality for absolute value



For any $u \in \mathbb{R}$ and any $u^{(t)} \neq 0$,

$$|u| \leq \frac{u^2}{2|u^{(t)}|} + \frac{|u^{(t)}|}{2}.$$

Why is this true?

$$0 \leq (|u| - |u^{(t)}|)^2 = u^2 - 2|u||u^{(t)}| + |u^{(t)}|^2.$$

Rearranging gives the inequality.

Equality holds when $|u| = |u^{(t)}|$, so in particular at the current residual.

Majorizing the LAD objective

Let

$$r_i(\beta) = y_i - \mathbf{x}_i^\top \beta, \quad r_i^{(t)} = r_i(\beta^{(t)}).$$

Using the inequality,

$$|r_i(\beta)| \leq \frac{r_i(\beta)^2}{2|r_i^{(t)}|} + \frac{|r_i^{(t)}|}{2}.$$

Therefore, a majorizer of the LAD objective is

$$g(\beta \mid \beta^{(t)}) = \sum_{i=1}^n \left[\frac{(y_i - x_i^\top \beta)^2}{2|r_i^{(t)}|} + \frac{|r_i^{(t)}|}{2} \right].$$

The second term does not depend on β .

The resulting weighted least-squares update



Minimizing the majorizer is equivalent to minimizing

$$\sum_{i=1}^n w_i^{(t)} (y_i - x_i^\top \beta)^2, \quad w_i^{(t)} = \frac{1}{2|r_i^{(t)}|}.$$

Thus

$$\boxed{\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} y,}$$

where

$$W^{(t)} = \text{diag}(w_1^{(t)}, \dots, w_n^{(t)}).$$

In practice, replace $|r_i^{(t)}|$ by $\sqrt{(r_i^{(t)})^2 + \varepsilon}$ or $|r_i^{(t)}| + \varepsilon$ to avoid division by zero.

A scalar example with a nonsmooth penalty



Consider the objective

$$f(\theta) = \frac{1}{2}(y - \theta)^2 + \lambda|\theta|, \quad \lambda > 0.$$

This is a one-parameter version of a penalized least-squares problem.

The absolute-value term is nondifferentiable at $\theta = 0$.

Using the same inequality,

$$|\theta| \leq \frac{\theta^2}{2|\theta^{(t)}|} + \frac{|\theta^{(t)}|}{2}.$$

Hence a majorizer is

$$g(\theta \mid \theta^{(t)}) = \frac{1}{2}(y - \theta)^2 + \lambda \left\{ \frac{\theta^2}{2|\theta^{(t)}|} + \frac{|\theta^{(t)}|}{2} \right\}.$$

Solving the surrogate problem



Ignoring constants not involving θ , minimize

$$g(\theta \mid \theta^{(t)}) = \frac{1}{2}(y - \theta)^2 + \frac{\lambda}{2|\theta^{(t)}|}\theta^2 + \text{constant}.$$

Differentiate:

$$\frac{\partial g}{\partial \theta} = -(y - \theta) + \frac{\lambda}{|\theta^{(t)}|}\theta.$$

Set equal to zero:

$$-y + \theta + \frac{\lambda}{|\theta^{(t)}|}\theta = 0.$$

Therefore,

$$\theta^{(t+1)} = \frac{|\theta^{(t)}|}{|\theta^{(t)}| + \lambda} y.$$

shrinks y toward zero.

$$\theta^{(t+1)} = \frac{|\theta^{(t)}|}{|\theta^{(t)}| + \lambda} y$$

shrinks y toward zero.

- ▶ If λ is large, the shrinkage is stronger.
- ▶ If $|\theta^{(t)}|$ is small, the shrinkage is stronger.
- ▶ The update arises from replacing a nonsmooth penalty by a quadratic surrogate.

This is not the most efficient way to solve the scalar LASSO problem, but it shows how MM turns nonsmooth optimization into repeated smooth optimization.

Likelihood maximization through minorization



Suppose we want to maximize a log-likelihood $\ell(\theta)$. A common MM strategy is:

1. At $\theta^{(t)}$, find a lower bound

$$q(\theta \mid \theta^{(t)}) \leq \ell(\theta).$$

2. Make sure the bound touches:

$$q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).$$

3. Maximize the lower bound:

$$\theta^{(t+1)} = \arg \max_{\theta} q(\theta \mid \theta^{(t)}).$$

This is a direct analogue of EM, but the lower bound may come from analytical inequalities rather than missing-data conditioning.

Quadratic minorization for concave likelihoods



For many GLM-type likelihoods, $\ell(\beta)$ is concave but not quadratic.

A quadratic minorizer has the form

$$q(\beta \mid \beta^{(t)}) = \ell(\beta^{(t)}) + \nabla \ell(\beta^{(t)})^\top (\beta - \beta^{(t)}) - \frac{M}{2} \|\beta - \beta^{(t)}\|^2,$$

where M is chosen large enough.

Maximizing this gives

$$\beta^{(t+1)} = \beta^{(t)} + \frac{1}{M} \nabla \ell(\beta^{(t)}).$$

For maximization, this is gradient ascent viewed as a minorization-maximization algorithm.

Comparison with Newton's method



Newton's method uses local curvature:

$$\theta^{(t+1)} = \theta^{(t)} - \left\{ \nabla^2 f(\theta^{(t)}) \right\}^{-1} \nabla f(\theta^{(t)}).$$

It can converge very fast near the optimum, but may fail to decrease f globally unless modified.

MM instead prioritizes monotonicity:

$$f(\theta^{(t+1)}) \leq f(\theta^{(t)}) \quad \text{or} \quad \ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).$$

Newton methods are often fast but need safeguards. MM methods are often stable but may be slow if the surrogate is too conservative.

What makes a good surrogate?



A good surrogate should satisfy two requirements.

1. **Computational simplicity:** optimizing it should be easier than optimizing the original objective.
2. The surrogate should not be too loose. A very loose surrogate may give tiny steps and slow convergence.

- ▶ A tight majorizer follows the original objective closely near $\theta^{(t)}$.
- ▶ A loose majorizer may guarantee descent but take overly cautious steps.
- ▶ A surrogate that is easy to optimize but extremely loose may be computationally inefficient.

This mirrors EM: EM is stable and monotone, but in some problems it converges slowly.

- ▶ small change in parameter:

- ▶ small relative change in objective:

- ▶ small norm of gradient or score;

Monotonic improvement does not by itself guarantee that the algorithm has reached the global optimum.

MM inherits the usual difficulties of nonconvex optimization.

- ▶ For nonconvex minimization, MM may converge to a local minimum or stationary point.
- ▶ For nonconcave likelihood maximization, MM may converge to a local maximum or saddle-type stationary point.
- ▶ Different starting values may lead to different solutions.

The monotonicity theorem says that each step improves the objective. It does not say that the final answer is globally best.

This point is familiar from EM for finite mixtures.

Exercise 1: verify a majorizer



Let

$$f(\theta) = |\theta|.$$

For a current value $\theta^{(t)} \neq 0$, define

$$g(\theta \mid \theta^{(t)}) = \frac{\theta^2}{2|\theta^{(t)}|} + \frac{|\theta^{(t)}|}{2}.$$

Show that g majorizes f at $\theta^{(t)}$. Then find the MM update for minimizing

$$\frac{1}{2}(y - \theta)^2 + \lambda|\theta|.$$

Exercise 2: minorization and ascent



Suppose $q(\theta \mid \theta^{(t)})$ satisfies

$$q(\theta \mid \theta^{(t)}) \leq \ell(\theta) \quad \text{for all } \theta,$$

and

$$q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).$$

Suppose $\theta^{(t+1)}$ only improves the surrogate:

$$q(\theta^{(t+1)} \mid \theta^{(t)}) \geq q(\theta^{(t)} \mid \theta^{(t)}),$$

but does not necessarily maximize it.

Does $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$ still follow?

Solution to Exercise 2



Yes. Exact maximization is not needed.

The same inequality chain works:

$$\ell(\theta^{(t+1)}) \geq q(\theta^{(t+1)} \mid \theta^{(t)})$$

by minorization.

By assumed improvement,

$$q(\theta^{(t+1)} \mid \theta^{(t)}) \geq q(\theta^{(t)} \mid \theta^{(t)}).$$

By tangency,

$$q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}).$$

Therefore,

$$\boxed{\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})}.$$

1. Start with $\theta^{(0)}$.

- $$g(\theta \mid \theta^{(t)}) \geq f(\theta), \quad g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)}).$$

$$g(\theta \mid \theta^{(t)}) \geq f(\theta), \quad g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)}).$$

- $$\theta^{(t+1)} = \arg \min_{\theta} g(\theta \mid \theta^{(t)}).$$

- For maximization, reverse the inequalities and maximize the minorizer.

Key comparisons



Method	Core idea	Strength
EM	conditional expectation	natural for missing data
MM	surrogate optimization	very general
Newton	local quadratic approximation	fast near optimum
Gradient descent	first-order quadratic bound	simple, scalable

Key Takeaway Messages



1. MM algorithms replace a hard objective by an easier surrogate.
2. For minimization, use a majorizer and minimize it.
3. For maximization, use a minorizer and maximize it.
4. The touching condition is essential for monotonicity.
5. EM is a special and important MM-type algorithm for incomplete-data likelihoods.
6. MM can also handle robust regression, penalized estimation, GLMs, and many other optimization problems.

Bound the difficult objective, optimize the easier bound, and inherit monotone improvement.

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